

Christian Giancola

MECHANICAL ENGINEERING · ROBOTIC CAM · ENGINEERING SOFTWARE

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PROFILE

Mechanical Engineering BSc candidate focused on robotic additive manufacturing, computational geometry and engineering software. I develop multi-axis toolpaths in Rhino and Grasshopper and Python postprocessors for KRL generation, coordinate transformations, tool orientation, extrusion control and external-axis coordination. I want to apply this combination of manufacturing knowledge and software development to automated CAD, CAM and robotic process-planning workflows.

EXPERIENCE

Industrial Robotics and Additive Manufacturing Intern | University of Bologna **2026 – Present**

Montecuccolino Laboratory, Bologna

- Developed software-driven toolpaths for robotic material deposition using a six-axis KUKA robot, Rhino 8, Grasshopper and KUKA|prc.
- Designed planar, inclined and non-planar slicing strategies for complex surfaces and multi-axis trajectories.
- Developed Python postprocessors that convert geometric path data into KRL programs and manage coordinate transformations, tool orientation, extrusion, an external linear axis, homing and purge sequences.
- Integrated, configured and tested a system comprising the robot, extrusion unit, tool and external linear axis.
- Built parametric geometry-processing workflows for curve ordering, path continuity, surface-normal control and local tool-frame generation.
- Simulated and checked reachability, singularities, orientation and trajectories before execution on the robot.
- Performed hands-on printing tests and iteratively modified geometry, process parameters and code based on observed results.

PROJECTS

KinePath | Personal software project · chrigia12.github.io/KinePath **2026**

- TypeScript web application that imports meshes and BREPs, generates planar and non-planar toolpaths, checks kinematics and collisions for the KUKA KR16 and exports KRL programs. Developed from laboratory process-planning requirements.

Toolpath Check | Personal software project **2026**

- Browser-based tool for analysing and validating robotic paths through geometric checks, program inspection and workflow automation.

FrameForge | Personal software project **2026**

- Browser-based platform for generating usable CAD geometry from images, videos and mechanical drawings.

EDUCATION

BSc in Mechanical Engineering | University of Bologna **2023 – Present**

- Experimental thesis, in progress, on the generation and control of toolpaths for robotic additive manufacturing.
- Main areas: mechanics, mechanical design, manufacturing technology, automation, programming and industrial systems.

First-level Academic Diploma Programme in Classical Guitar | G. B. Martini Conservatory, Bologna **2024 – Present**

TECHNICAL SKILLS

Automation and robotics: KUKA KRL, KUKA|prc, offline programming, simulation, six-axis robots, external axes

Robotic CAM: robotic deposition, planar and non-planar slicing, multi-axis toolpaths, tool orientation, collision-aware planning, technical validation

Software development: Python, TypeScript, C++, Three.js, postprocessing, geometric-data processing, automated testing, browser-based engineering tools

Computational design: Rhino 8, Grasshopper, curves, surfaces, BREPs, meshes, data trees, computational geometry

CAD and drafting: PTC Creo, AutoCAD, 3D modelling, technical drawings

Engineering methods: prototyping, machine testing, troubleshooting, technical documentation, iterative improvement

LANGUAGES AND ADDITIONAL INFORMATION

Italian: native · **English:** B2, Cambridge First Certificate (Grade C), used for technical documentation and communication

Italian and EU citizen · Driving licence B · Basketball referee · Available from February 2027 for short technical placements and engineering software projects